

1. (10 points) Consider the following directed graph:



- (a) (2 points) Write the joint distribution $P(x_1, x_2, x_3, x_4)$ in terms of a product of conditional distributions, one for each node in the graph.
- (b) (4 points) A possible application of this directed graph is to model a stimulus that changes over time. Let

$$x_1 \sim \mathcal{N}(0, \sigma^2)$$

$$x_t | x_{t-1} \sim \mathcal{N}(x_{t-1}, \sigma^2),$$

where $t = 2, 3, 4$. This is known as a *random walk model*, since x_t is obtained by adding a random increment (in this case, a Gaussian) to x_{t-1} . What is the 4×4 covariance matrix of the vector $[x_1 \ x_2 \ x_3 \ x_4]^T$ in terms of σ^2 ?

- (c) (2 points) Find the inverse of the covariance matrix found in part (b) in terms of σ^2 . This is known as the *precision matrix*.
- (d) (2 points) Relate where zeros appear in the precision matrix to the graph structure.

a) For a directed graph, it is a product of each.

$$x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_4$$

$$\Rightarrow P(x_1, x_2, x_3, x_4) = P(x_1) \cdot P(x_2 | x_1) \cdot P(x_3 | x_2) \cdot P(x_4 | x_3)$$

b) $x_1 \sim \mathcal{N}(0, \sigma^2)$

$$x_t = x_{t-1} + \xi_t, \text{ where } \xi_t \sim \mathcal{N}(0, \sigma^2) \text{ independently}$$

$$\Rightarrow x_1 = \xi_1$$

$$x_2 = \xi_1 + \xi_2$$

$$x_3 = \xi_1 + \xi_2 + \xi_3$$

$$x_4 = \xi_1 + \xi_2 + \xi_3 + \xi_4$$

$$\text{Cov}(x_i, x_j) = \min(i, j) \cdot \sigma^2 \quad (\text{since the } \xi_k \text{ are independent with variance } \sigma^2).$$

$$\Rightarrow$$

$$\Sigma = \sigma^2 \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

c) $M = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 2 & 2 & 0 & 1 & 0 & 0 \\ 1 & 2 & 3 & 3 & 0 & 0 & 1 & 0 \\ 1 & 2 & 3 & 4 & 0 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 2 & -1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 3 & -1 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & -1 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 0 & 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 & -1 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 2 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array} \right]$$

$$\Sigma^{-1} = \frac{1}{\sigma^2} \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

d) The zeros in the precision matrix correspond exactly to pairs of nodes that are not directly connected.

$x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_4$, the directed edges are only between consecutive nodes.

The precision matrix $\Lambda_{ij} = 0$, whenever $|i-j| > 1$, $\Rightarrow$

$\Lambda_{13} = \Lambda_{31} = 0 \rightarrow x_1$ and x_3 are not directly connected

$\Lambda_{14} = \Lambda_{41} = 0 \rightarrow x_1$ and x_4 are not directly connected

$\Lambda_{24} = \Lambda_{42} = 0 \rightarrow x_2$ and x_4 are not directly connected

This is a general principle for Gaussian Markov Random fields.

The (i,j) entry of the precision matrix is zero iff x_i

and x_j are conditionally independent given all other variables,

which in a graphical model corresponds to having no

direct edge between them. The precision matrix is essentially

the adjacency structure of the graph (plus the diagonal).